NOT MEASUREMENT SENSITIVE

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23 February 1996
SUPERSEDING
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2 December 1994

DETAIL SPECIFICATION

MEDALLION, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for medallions used by the Department of Defense and other Government Agencies (see 6.1).

1.2 Classification. The medallions covered by this specification shall be as specified on the applicable military specification sheet (See 2.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION SHEETS

(See Supplement 1 for list of specification sheets)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be addressed to: Director, The Institute of Heraldry, US Army, 9325 Gunston Road, Room S112, Fort Belvoir, VA 22060-5579, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 8455

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.



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(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 42, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DoDISS - Department of Defense Index of Specifications and Standards

(Copies of the DoDISS are available on a yearly subscription basis either from the Government Printing Office or the DoDSSP Subscription Services, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

STANDARD METAL CHIP SET

THE INSTITUTE OF HERALDRY, U.S. ARMY STANDARD METAL FINISH CHIP SET

(The standard color chip(s) for finish may be obtained from the procuring activity for Government procurements.)

2.3 Non-Government publications. The following documents form a part of this document to the extend specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B-487 - Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Measurement of

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern. The design of the medallion shown on the specification sheet are for illustrative purposes only.



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3.2 First article. When specified in the contract or purchase order (see 6.2) a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Recycled, recovered or environmentally preferable materials. Recycled, recovered, or environmental preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials. Materials shall as specified herein. Recovered material shall be used to the maximum extent possible.

3.4.1 Copper base alloy. The copper base alloy shall be roll polished, free from pits, scale (including red oxide), dents, nicks, cracks, scratches, segregation and foreign inclusions that will not be removed in later processing. When tested as specified in 4.6.1, the chemical composition of the copper base alloy shall be a specified in Table I.

TABLE I. Chemical composition of copper base alloys.

| ALLOY | COPPER | LEAD (MAX.) | IRON (MAX.) | OTHERS |
|---|---|---|---|---|
| Red Brass | 84.0-86.0 | 0.05 | 0.05 | Remainder |
| Bronze | 89.0-91.0 | 0.05 | 0.05 | Remainder |

3.4.2 Gold for plating. The gold for plating shall be 24 karat.

3.4.3 Silver for plating. Silver for plating shall be not less than 99.0 percent pure silver.

3.4.4 Nickel for plating. Nickel for plating shall be the nickel plating normally used in commercial practice.

3.4.5 Lacquer. Lacquer shall be a pale clear synthetic lacquer. The use of a pigmented lacquer shall not be permitted.

3.5 Design. The embossed design of each medallion shall be an exact duplicate of the design on the Government loaned hubs (see 3.10), from which the contractor's working dies shall be extracted. The contractor's working die shall be tooled and polished to remove any dents, nicks, scratches or other imperfections.

3.6 Construction.

3.6.1 Stamping and trimming. The stamping shall have a well defined die struck edge. The medallion shall be trimmed to the die struck edge. All edges shall be clean, smooth and free from burrs, drag, step and rough edges. The stamping and trimming operations shall not damage or distort the design or alter the shape of the item.

3.7 Dimensions. All dimensions shall be as indicated. When referenced dimensions are given, these dimensions are furnished as information for bidding purposes only. The actual dimensions will be controlled by the Government loaned hub.

3.8 Finish. Unless otherwise specified, medallions shall be finished as specified on the applicable specification sheet and shall match the color and finish of the standard metal finish chip. Where



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plating or oxidizing is specified, the entire item (front, back, and edges) shall be plated or oxidized as applicable. In addition to the finish specified on the specification sheets, all silver, rose gold plated and unplated red brass or bronze medallions shall be completely coated with lacquer specified in 3.9. Gold plated medallions may be lacquered at the option of the manufacturer.

3.8.1 Plating. All plating shall be by electroplating methods. Plating shall be non-porous and continuous over the entire plated surface. There shall be no cut-through, peeled or blistered plating. The plating shall be smooth, fine grained, adherent and free from pits, nodules, porosity, indications of burning, excessive edge build up and other detrimental defects.

3.8.1.1 Gold plating. The gold plating shall be accomplished using gold specified in 3.4.2. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating for gold shall not be permitted. When tested as specified in 4.6.2, no visible chemical reaction shall appear such as evolution of gases.

3.8.1.2 Nickel plating. Nickel for plating shall be not less than 0.0003 inch (0.0008 cm) thick. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487.

3.8.1.3 Nickel underplating. When nickel underplating is specified on the applicable specification sheet, the underplating shall be not less than 0.00025 inch (0.000635 cm) thick. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487.

3.8.1.4 Silver plating. When silver plating is specified, silver specified in 3.4.3 shall be used for plating. After finishing, the silver plating shall be not less than 0.0003 inch (0.0008 cm) thick. Items made of brass shall be silver thickness tested on their obverse side. In case of a dispute, testing shall be conducted in accordance with ASTM-B-487.

3.8.2 Cutting down. When necessary to remove nicks, scratches, pin holes or other blemishes, the medallion shall be cut down by means of grease or oil buffing. The medallion shall not be cut down to the extent that any details of the design are obliterated or reduced.

3.9 Lacquering. Gold alloy or gold plated parts may be lacquered at the option of the manufacturer; however, all rose gold items shall be lacquered. The dry lacquer film shall be hard, continuous, level, adherent and free from lint, dust and other foreign matter. When tested as specified in 4.6.3.1, there shall be no change in the appearance of the clear lacquer surface except a slight overall yellowing of the highlights. When tested as specified in 4.6.3.2, the tissue paper shall not adhere to the clear lacquered surface.

3.10 Government loaned property. Hubs will be loaned by the Government and shall be used to make the contractor's working dies necessary for one contract or order (see 3.4).

3.11 Marking for identification. In addition to any special marking required by the contract or purchase order (See 6.2), the contractor shall stamp his trade or other identifying mark and the letters "GI" (GI for government procurement only) legibly and inconspicuously on the periphery at the bottom of the medallion. The marks shall not be more than 1/16 inch (0.16 cm) in height.



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3.12 Marking for cleaning. Marking for cleaning of gold plated items shall be as follows: "To clean, rub gently with a soft, dry cloth. Do not clean with a chemically treated cloth or abrasive".

3.13 Workmanship. The finished item shall be clean, well made and shall conform to the acceptable quality levels established by this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in 4.5 and 4.6.

4.3 First article inspection. Inspection and testing of the first article (see 3.2) shall be made of a completely finished item for all provisions of this specification applicable to the end product examination and tests.

4.4 Conformance inspection. Inspection shall be in accordance with the provisions set forth herein, except where otherwise indicated (see 6.3).

4.4.1 Conformance inspection sampling.

4.4.1.1 Testing of components. In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on components and materials listed in Table II for the test characteristics shown therein.

Table II. Testing of components.

| COMPONENT & LOT EXPRESSED IN TERMS OF | CHARACTERISTIC | RQM'T PARA | TEST METHOD | RQM'T APPLICABLE TO LOT AVERAGE | # DETERM PER SAMPLE UNIT | RESULTS REPORTED AS | SAMPLE UNIT |
|---|---|---|---|---|---|---|---|
| *Copper base allow (1 LB) as applicable | Chemical composition | 3.4.1 | 4.6.1 | X | 2 composite | Nearest 0.1% for each | 4 oz of composite |
| *Gold used for plating (1 oz) | Karat | 3.4.2 | STD comm'l | X | 1 composite | Nearest 0.1%n | 1/2 oz |
| *Silver for Plating | Fineness | 3.4.3 | STD comm'l | X | 2 composite | Nearest 0.1% | 1 oz |

*Material so indicated may be accepted on a contractor's certificate of compliance, based on the quality markings on the medallions. The Government reserves the right to test samples of the gold and silver (without additional charge) to verify the contractor's certification.

4.5.1.1.1 Certification of compliance. Materials listed below may be accepted on the basis of the contractor's certificate of compliance for requirements specified in applicable paragraphs of this specification.



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| COMPONENTS | CHARACTERISTICS | RQM'T PARA |
|---|---|---|
| Copper base alloy | Material identification | 3.4.1 |
| Gold for plating | Material identification & karat | 3.4.2 |
| Silver for plating | Material identification & fineness | 3.4.3 |
| Nickel for plating | Material identification | 3.4.4 |
| Lacquer | Material identification | 3.4.5 |

4.5.1.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplished of the following is in accordance with the specification requirements.

4.5.2 Inspection of the end item.

4.5.2.1 Visual examination of medallions. Examination shall be made at a distance of approximately 16 to 22 inches with illumination equal to average daylight and arranged so as to avoid as much reflected light as possible. The defects found during examination shall be classified in accordance with 4.5.2.1.1. The lots shall be inspected in accordance with 4.5.2.1.2. The unit of product for the examinations in 4.5.2.1.1 shall be one completely fabricated medallion.

4.5.2.1.1 Examination of medallions for defects in finish, design, material, construction, workmanship, and marking. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

TABLE III. Defects.

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | MAJOR | (*) | MINOR |
| FRONT: | | | | |
| Color & Finish | Poor match to the standard or approved sample | X | | |
| | Does not match the standard or approved sample as follows: | | | |
| | The oxidized finish, when required, has not been sufficiently relieved, or has been excessively relieved | | * | |
| | Is not highlighted to the same extent as the standard or approved sample | | * | |
| | Background or recessed areas are not finished as specified | X | | |
| | Surface contains pits, scale, nick, scratch, crack, machine mark, pin hole, segregation, rupture, foreign matter or other blemish | | * | |



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TABLE III. Defects (Continued).

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | MAJOR | (*) | MINOR |
| | Discoloration, spot, stain or speck affecting appearance | | | X |
| | Buff drag, blushed, or cloudy finish | | * | |
| | Discolored, i.e., spot, stain, or speck | | * | |
| | Not clean | | * | |
| | | | | |
| Plating | Not plated (when required) | X | | |
| | Not type specified | X | | |
| | Plating is not smooth, continuous or adherent | X | | |
| | Cut through, porous, pitted, crystallized flaking, peeling or blistered | X | | |
| | | | | |
| BACK & EDGES: | | | | |
| Color & finish | Surface contains pits, scale, nick, scratch, crack, machine mark, pin hole, rupture, segregation or other blemish | | | X |
| | Foreign matter embedded | | | X |
| | Discoloration, spot, stain or speck affecting appearance | | | X |
| | Buff drag, blushed, or cloudy finish | | | X |
| | Not clean | | * | |
| | | | | |
| Plating | Not plated (when required) | X | | |
| | Not type specified | X | | |
| | Plating is not smooth, continuous or adherent | X | | |
| | Cut through, porous, pitted, crystallized flaking, peeling or blistered | X | | |
| | | | | |
| GENERAL: | | | | |
| Construction | Metal marks on exposed surface, such as nick, dent, dig, gouge or scratch | | * | |
| | Not trimmed to die-struck edge | | * | |
| | Operations are poorly accomplished | | * | |
| | Letters distorted | | * | |
| | Surface or edge not clean, not smooth or not free from burrs, roughness, drag, step tool marks | X | | |
| | Metal marks on exposed surface, such as nick, dent, dig, gouge or scratch | | * | |
| | | | * | |
| Lacquer | Not lacquered when required | X | | |
| | Areas of no lacquer | | * | |
| | Lacquer forms noticeable runs or sharp coarse particles | | * | |
| | Foreign matter embedded in finish, i.e., lint, dust, etc. | | * | |



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TABLE III. Defects (Continued).

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | MAJOR | (*) | MINOR |
| | Hazy, rainbow effect, cloudy, or powdering | | * | |
| | Not smooth, continuous or adherent, i.e., flaking, blistering, or peeling | | * | |
| | Not set to touch, i.e., tacky when pressure is applied to coating | | | X |
| | | | | |
| Design | Details altered, does not conform to Government hub, drawing, or standard sample | X | | |
| | Any detail struck over resulting in double impression | X | | |
| | Any operation not in accordance with the specified requirements | | * | |
| | Any warp, twist, or distortion producing irregular surface contour or outline | | * | |
| | Any significant detail not clear, marred, missing, reduced or obliterated, i.e., does not compare favorably with approved sample. | X | | |
| | Design not as specified | X | | |
| | | | | |
| Quality of metal | Surface pitted, porous, crystalline, spotted or opened grained | X | | |
| | | | | |
| Workmanship | Back not flat or smooth when required | | | X |
| | Any sharp or rough edge, burr, drag, or step | | * | |
| | | | | |
| Workmanship | Missing, incorrect, illegible, misspelled, not accomplished as specified, or not placed as required | | | X |

4.5.2.1.2 Examination of decorations for defects in dimensions. Any dimension which is not within the specified tolerance shall be classified as a defect.

4.5.2.2 End item testing. Testing of the completely fabricated medallion shall be performed in accordance with Table IV for the characteristics shown therein. The sample unit shall be one completely fabricated medallion. The results shall be reported as pass or fail for the examination.



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TABLE IV. Testing of end item.

| CHARACTERISTIC | REQUIREMENT PARAGRAPH | TEST METHOD | # DETERM PER SAMPLE /UNIT | RESULTS REPORTED AS |
|---|---|---|---|---|
| | | | | |
| Test For Copper Base Alloy | 3.3.1 | 4.6.1 | 1 | Pass or Fail |
| | | | | |
| Tests For Plating | | | | |
| Gold Plating | 3.7.1.1 | STD comm'l | 1 | Pass or Fail |
| Nickel Plating | 3.7.1.2 | STD comm'l | 1 | Pass or Fail |
| Nickel Plating Thickness | 3.7.1.2 | STD comm'l | 1 | Pass or Fail |
| Nickel Underplating | 3.7.1.3 | STD comm'l | 1 | Pass or Fail |
| Silver Plating Thickness | 3.7.1.4 | STD comm'l | 1 | Pass or Fail |
| | | | | |
| Acid Test: | | | | |
| Gold Plating | 3.7.1.1 | 4.6.2 | 1 | Pass or Fail |
| | | | | |
| Tests For Lacquer: | | | | |
| Liver Of Sulfur | 3.8 | 4.6.3.1 | 1 | Pass or Fail |
| Tackiness Test | 3.8 | 4.6.3.2 | 1 | Pass or Fail |

4.6 Tests.

4.6.1 Chemical composition test for copper base alloy. Chemical composition shall be determined by standard commercial testing. Results shall be evaluated to determine compliance with the requirements specified in 3.3.1.

4.6.2 Acid test for gold plating. If the test insignia has been lacquered, the lacquer shall be completely removed prior to testing with an organic solvent. The test acid shall be applied as follows: Place a drop of acid of not less than 1/16 inch (0.16 cm) in diameter on three (3) different spots on the plated surface allowing the drops to remain for not less than one (1) minute during which time the surface of the item shall be inspected to determine compliance with the requirements in 3.7.1.1. No visible chemical reactions shall occur. Acid drops to be placed on flat surfaces on the obverse or reverse face when possible. A minimum of two spots must withstand the acid test. The test acid shall be applied at room temperature (60 to 80 degrees Fahrenheit/15.6 to 26.7 degrees Celsius) and shall consist of a solution containing 50 percent by volume of chemically pure nitric acid (specific gravity 1.42) and an equal volume of distilled water.

4.6.3 Test for lacquer.

4.6.3.1 Liver of sulfur test. Items to be tested shall be immersed in a 2 percent by weight chemically pure liver of sulfur (potassium sulfide) water solution at a temperature of 100 degrees Fahrenheit ± 10 degrees Fahrenheit (37.8 degrees Celsius ± 12.2 degrees Celsius) for a period of 3 minutes. The item shall then be removed and rinsed in warm, then cold, then hot water. The item shall be wiped gently with an absorbent cellulose material or whirled to remove residual moisture, allowed to dry at room temperature (60 to 80 degrees Fahrenheit/15.6 to 26.7 degrees Celsius) for one (1) hour and then be examined to determined compliance with 3.9.



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4.6.3.2 Tackiness test. At room temperature (60 to 80 degrees Fahrenheit/15.6 to 26.7 degrees Celsius), press a piece of tissue paper against the lacquered surface for 15 seconds, using any pressure capable of being exerted between thumb and two fingers, after which the pressure shall be released and the item inspected to determine compliance with 3.9.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Medallions covered by this specification are intended to be presented to government personnel to provide tangible evidence of public recognition of personal acts of heroism performed or valuable services rendered.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. The number and date of the applicable specification sheet (see 1.2 and 2.1).

d. When a first article is not required (see 3.2).

e. Packaging requirements (see 5.1).

f. When special marking is required (see 3.11 & 3.12).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed item. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.



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6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.10.

6.5 Subject term (key word) listing.

Awards
Medals

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - IH
Air Force - 45
Navy - NU

Review activities:
Air Force - 32, 82, 99
Navy - MC
Coast Guard - CG
DLA - CT

Preparing activity:
Army - IH

(Project No. 8455-0827)



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